Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

GTIIT

Calculus 1 - M
Winter Semester - 2024
Lecture 15 - December 2nd

1st Part

- The Fundamental Theorem of Calculus (Section 5.3)
- The substitution Rule (Section 5.5).

Substitution Rule for Antiderivatives

Given any antiderivative F of f,

$$\int f(g(x)) \cdot g'(x)\, dx = F(g(x)) + C$$

As a consequence of the Chain Rule for Derivatives, denote by $u = g(x) \Rightarrow \frac{du}{dx} = g'(x)$ and so, we obtain:

$$\boxed{\int f(g(x)) \cdot g'(x)\, dx = \int f(u)\, du = F(g(x)) + C \checkmark}$$

This means: It is permissible to operate with dx and du after integral signs as if they were differentials.

Example: Solve $\int \frac{6}{2+3x^2}dx$.

Solution. • Relate the sum of tue squares in the denominator with $\boxed{\int \frac{1}{1+u^2}du = \arctan(u) + C.}$

• So, let us rewrite the given integral:

$$\int \frac{6}{2+3x^2}dx = \int \frac{3\cdot 2}{2\cdot\left[1+\left(\sqrt{\frac{3}{2}}x\right)^2\right]}dx = 3\int \frac{1}{1+\left(\sqrt{\frac{3}{2}}x\right)^2}dx$$

• Denote $u = \sqrt{\frac{3}{2}}x \Rightarrow \frac{du}{dx} = \sqrt{\frac{3}{2}}$ and substitute in the integral:

$$\int \frac{6}{2+3x^2}dx = 3\int \frac{1}{1+\left(\sqrt{\frac{3}{2}}x\right)^2}dx = 3\int \frac{1}{1+u^2}\sqrt{\frac{2}{3}}du = \sqrt{6}\arctan(u)+C$$

$$\boxed{= \sqrt{6}\arctan\left(\sqrt{\frac{3}{2}}x\right) + C \checkmark}$$

The Fundamental Theorem of Calculus

Estimation of the definite integral using absolute minimum and maximum values of f

Lemma: Suppose that $m \leq f(x) \leq M$ for any $x \in [a, b]$, then the definite integral of f on $[a, b]$ is abounded by

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$$

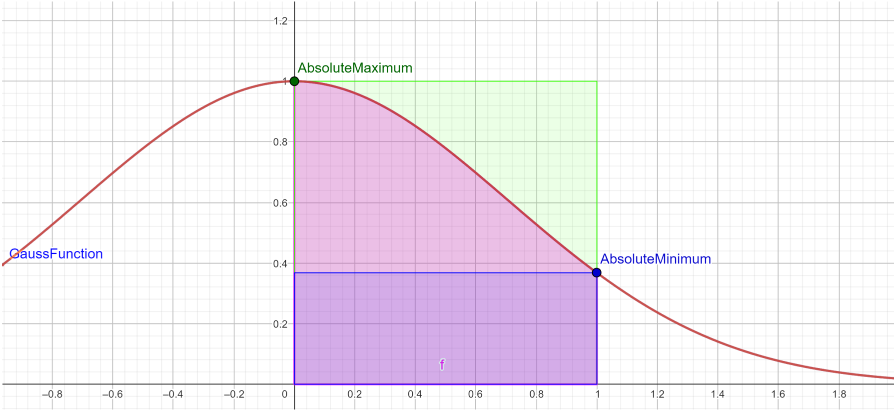
Example 1. a) Find the absolute minimum and maximum values of $f(x) = e^{-x^2}$ on $[0, 1]$.

b) Use part a) to estimate the value of $\int_0^1 e^{-x^2}\,dx$.

- The function e^{-x^2} has no antiderivative and can only be numerically computed, by approximations.

The Gauss function of extended use in Probability Theory

Area under the curve $f(x) = e^{-x^2}$ on $[0, 1]$



The Gauss function used in Probability

Solution.

- Since f is the composition of e^u with $u = -x^2$, then $f(x) = e^{-x^2}$ is differentiable on $(0,1)$, and

$$f'(x) = -e^{-x^2} \cdot 2x < 0 \; on \; (0,1)$$

then, e^{-x^2} is decreasing on $[0,1]$

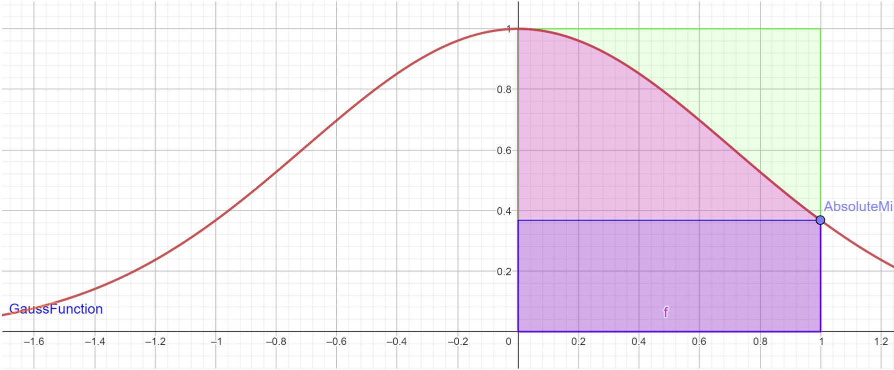
- $f(x) = e^{-x^2}$ on $[0,1]$ attains the
  1. absolute maximum on the left endpoint $(0,M)=(0,1)$
  2. absolute minimum on the right endpoint $(1,m)=(1,e^{-1})$

Hence, the property $m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$ says:

$$\boxed{0.37 \approx e^{-1} \leq \int_0^1 e^{-x^2}\,dx \leq 1 \checkmark}$$

The Gauss function of extended use in Probability Theory

Area under the curve $f(x) = e^{-x^2}$ on $[0, 1]$



$$0.37 \approx e^{-1} \leq \int_0^1 e^{-x^2}\,dx \leq 1$$

The Gauss function of extended use in Probability Theory

$f(x) = e^{-x^2}$

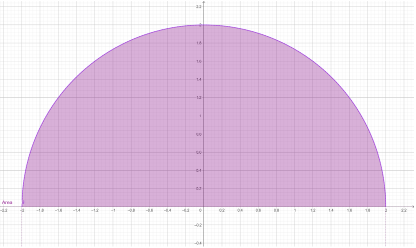
$f(x) = e^{-x^2} \quad on \quad [0, 1]$

$$0.37 \approx e^{-1} \leq \int_0^1 e^{-x^2}\, dx \leq 1$$

Example 2. Evaluate the following integrals by interpreting each in terms of areas. Sketch the graphs.

a) $\int_{-2}^{2}\sqrt{4-x^2}\,dx$ b) $\int_0^3 (x-1)\,dx$

Solution. a) $\int_{-2}^{2}\sqrt{4-x^2}\,dx$ is the area under the curve $y=\sqrt{4-x^2}$ on $[-2,2]$

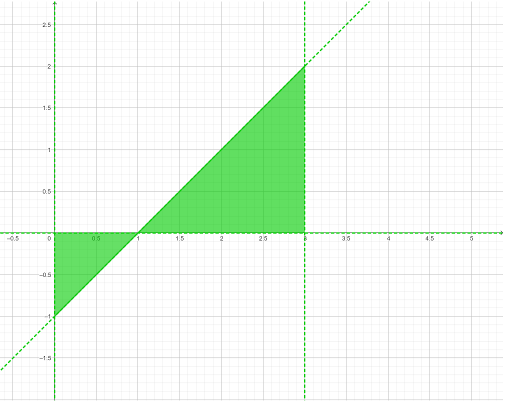


So, $\int_{-2}^{2}\sqrt{4-x^2}\,dx$= half of the area the circle $x^2+y^2=4$, centered at $(0,0)$ of radius $R=2$

$$\boxed{\int_{-2}^{2}\sqrt{4-x^2}\,dx=\frac{1}{2}\pi R^2} = \frac{1}{2}\pi 2^2 = 2\pi \checkmark$$

Example 2. b) $\int_0^3 (x-1)\,dx$

Solution. b) Integral = negative of the area of the triangle on the left plus the area of the triangle on the right



$$\Rightarrow \boxed{\int_0^3 (x-1)\,dx = -\frac{1\times 1}{2} + \frac{2\times 2}{2} = \frac{3}{2} \checkmark}$$

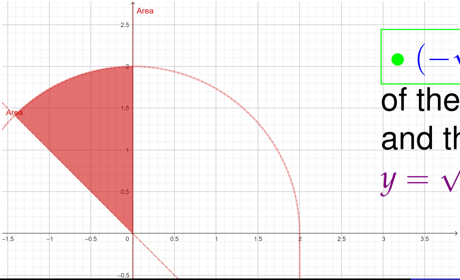
Example 2. c) Evaluate the following integral by interpreting it in terms of areas:

$$\int_{-\sqrt{2}}^{0}(\sqrt{4-x^2}+x)\,dx+\int_0^2\sqrt{4-x^2}$$

Solution. The 1st integral is the

area between the curves $y=\sqrt{4-x^2}$ and $y=-x$

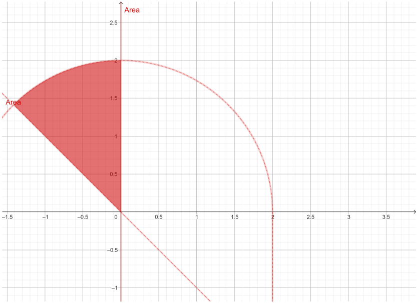
$$\int_{-\sqrt{2}}^{0}(\sqrt{4-x^2}+x)\,dx=\int_{-\sqrt{2}}^{0}(\sqrt{4-x^2}-(-x))\,dx$$



- $(-\sqrt{2},\sqrt{2})$ = intersection

of the line $y=-x$
and the cicrcumference
$y=\sqrt{4-x^2}$

Then the 1st integral: area between the two curves

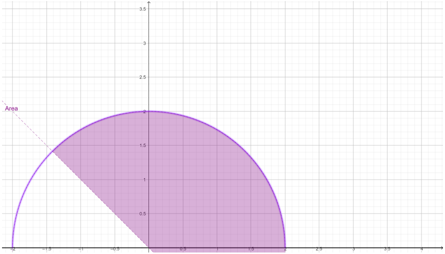


= $\frac{1}{8}$ of the area of the circle centered at $(0,0)$ of radius $R = 2$:

$$\boxed{\Rightarrow \int_{-\sqrt{2}}^{0}(\sqrt{4-x^2}+x)\,dx = \frac{1}{8}\pi R^2 \quad = \frac{1}{8}\pi 2^2 = \frac{1}{2}\pi \checkmark}$$

The area in terms of integrals

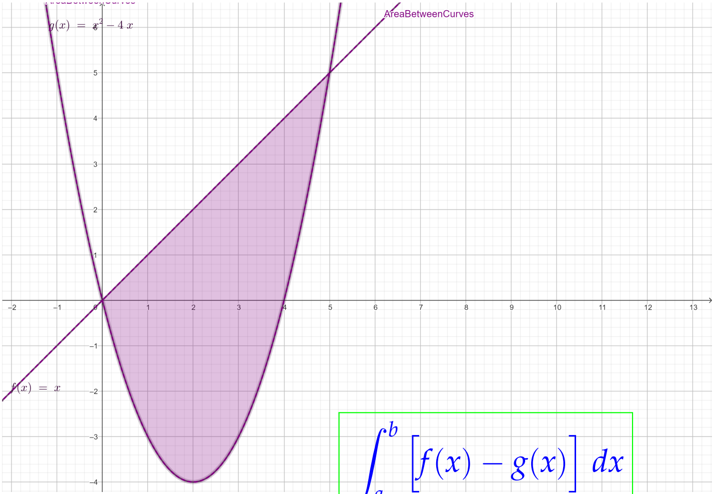
For the whole area:



= area of $\frac{3}{8}$ of the circle centered at $(0,0)$ of radius $R = 2$

$$\boxed{\int_{-\sqrt{2}}^{0} (\sqrt{4-x^2} + x)\, dx + \int_0^2 \sqrt{4-x^2}\, dx = \frac{3}{8}\pi R^2 \quad = \frac{3}{8}\pi 2^2 = \frac{3}{2}\pi \ \checkmark}$$

Area enclosed by two curves



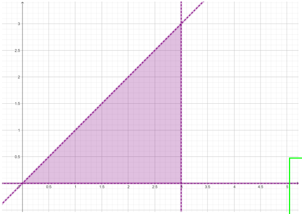
$$\int_a^b \left[f(x) - g(x) \right] \, dx$$

=area enclosed by the two curves $y = f(x)$ and $y = g(x)$

The Fundamental Theorem of Calculus

The integral function is an antiderivative of f

Curve $y = t$ from $t = 0$ to $t = x$:



$t = 0$ $\qquad$ $t = x$

$$\boxed{\text{Area of the triangle} = \frac{1}{2}\ \text{Base} \times \text{height}}$$

$$\boxed{\int_0^x t\,dt = \frac{1}{2}x^2}$$

- The integral from $c = 0$ to x is an antiderivative of $f(x)$:

$$\frac{d}{dx}\left[\int_0^x t\,dt\right] = x$$

The Fundamental Theorem of Calculus: Example 3.

- $\frac{d}{dx}\left[\int_0^x \cos(2t)\, dt\right] = \cos(2x)$

- $\frac{d}{dx}\left[\int_x^4 \sqrt{3t-1}\, dt\right] = \frac{d}{dx}\left[-\int_4^x \sqrt{3t-1}\, dt\right] = -\sqrt{3x-1}$

- $\frac{d}{dx}\left[\int_x^\pi \tan(t)\, dt\right] = \frac{d}{dx}\left[-\int_\pi^x \tan(t)\, dt\right] = -\tan(x)$

- $\frac{d}{dx}\left[\int_0^x \ln(3t+1)\, dt\right] = \ln(3x+1)$

The Fundamental Theorem of Calculus

Let f be continuous on $[a, b]$, then

1. The integral function

   $\boxed{F(x) = \int_a^x f(t)\, dt \text{ is s continuous on } [a, b]}$ and

   $\boxed{F \text{ is an antiderivative of } f \text{ on } (a, b).}$

   Equivalently, $\boxed{\dfrac{d}{dx} \int_a^x f(t)\, dt = f(x) \;\; on \;\; (a, b)}$.

2. If F is any antiderivative, that is, $F' = f$,

   $$\boxed{\int_a^b f(x)\, dx = F(b) - F(a)}.$$

3. $\boxed{\int_a^x F'(t)\, dt = F(x) - F(a) \text{ for any differentiable } F \text{ on } (a, b)}$

4. $F(x) = \int f(x)\, dx$ means $F'(x) = f(x)$ on an interval

True or false?

$$\boxed{\text{Is it true } \int_{-1}^{2} \frac{1}{x^2}\,\mathrm{dx} = -\frac{1}{x}\Big|_{x=-1}^{x=2} = -\frac{1}{2} - 1 = -\frac{3}{2}\ \textbf{?}}$$

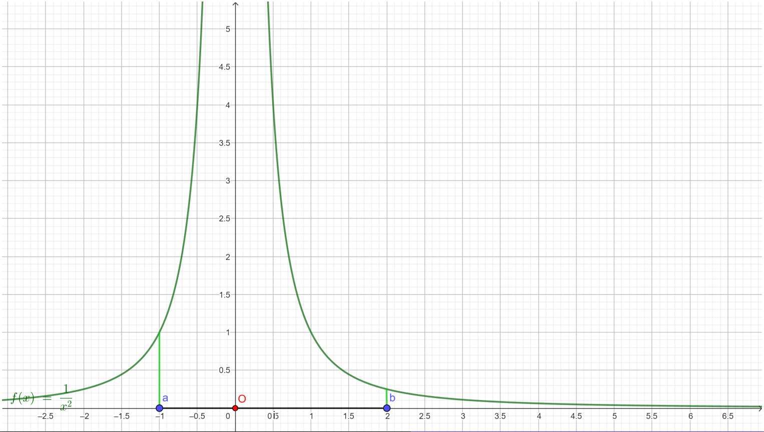
- It can not be true, since $\boxed{f(x) = \frac{1}{x^2} \geq 0 \Rightarrow \int_{-1}^{2} \frac{1}{x^2} dx \geq 0.}$
  - Moreover, the integral of $f(x) = \frac{1}{x^2}$ on $[-1, 2]$ does not make sense, since $x = 0 \notin \text{Dom}(\frac{1}{x^2})$, so, f is not continuous on $[-1, 2]$
  - Hence, the integral is not defined:

$$\cancel{\int_{-1}^{2} \frac{1}{x^2}\,\mathrm{dx} = -\frac{1}{x}\Big|_{x=-1}^{x=2}}$$

- The integral on $[-1, 2]$ is not defined because

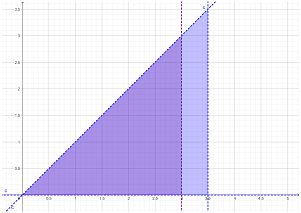
$x = 0 \notin \text{Dom}(\frac{1}{x^2})$

$$\cancel{\int_{-1}^{2} \frac{1}{x^2}\,\mathrm{dx} = -\frac{1}{x}\Big|_{x=-1}^{x=2}}$$



The Fundamental Theorem of Calculus

Proof. Let $\boxed{F(x) = \int_a^x f(t)\,dt}$, then integrate on $[a, x+h]$ and on $[a, x]$ and consider the difference of them:



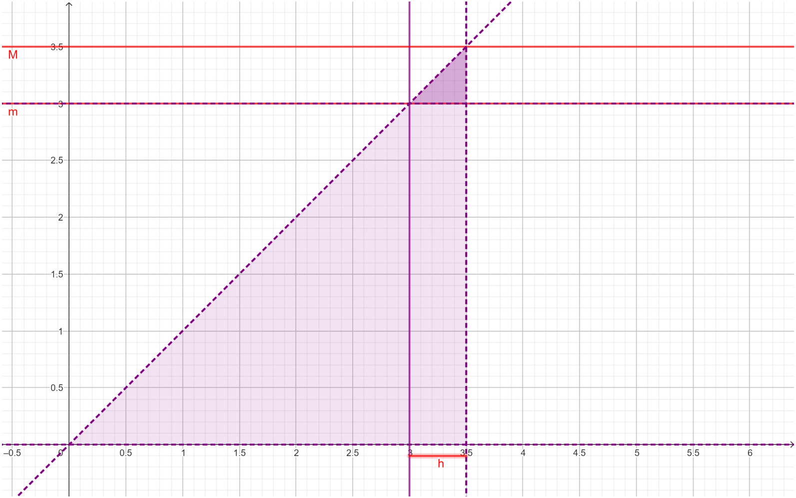
$< - - - - - - - - - >$ $[0, x]$

$< - - - - - - - - - - - - >$ $[0, x+h]$

$$\boxed{F(x+h) - F(x) = \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt = \int_x^{x+h} f(t)\,dt}$$

$$\Rightarrow \boxed{\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t)\,dt}$$

Since f is continuous on $[x, x+h]$, there exists $m = f(x_m)$ and $M = f(x_M)$ the absolute maximum and minimum:



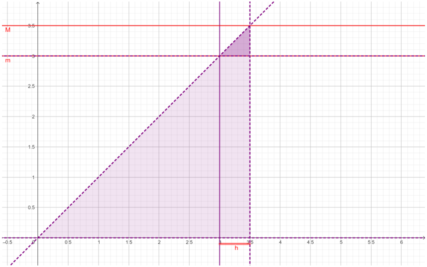
The area of the rectangle of height m is less than the integral of f, which is less than area of the rectangle of height M

Compare the integral with the areas of the rectangles of height m and M:

$$\boxed{m = f(x_m) \leq f(t) \leq f(x_M) = M}$$

$$\boxed{f(x_m)\cdot h \leq \int_x^{x+h} f(t)\, dt \leq f(x_M)\cdot h} \quad \textit{then}$$

$$\Rightarrow \boxed{f(x_m) \leq \frac{F(x+h)-F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\, dt \leq f(x_M)}$$

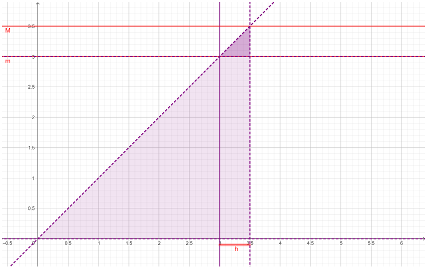


The area of the rectangle of height $f(m)$ is less than the integral of f, which is less than area of the rectangle of

The Fundamental Theorem of Calculus: $F'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x)$

Since f is continuous, by the Squeeze's Theorem:

$$\begin{array}{ccccc} f(x_m) & \le & \dfrac{F(x+h)-F(x)}{h} & \le & f(x_M) \\ \downarrow & & \downarrow & & \downarrow \\ f(x) & = & F'(x) & = & f(x)\ \checkmark \end{array}$$



The Fundamental Theorem of Calculus, part 2:
If F is any antiderivative of a continuous f on $[a, b]$ then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Proof. Let $G(x) = \int_a^x f(t)dt \Rightarrow$(part 1)$\Rightarrow G'(x) = f(x) \forall x \in (a, b)$

- If F is any other continuous function on $[a, b]$, which is an antiderivative of f on (a, b), then F and G differ by a constant: $\boxed{F(x) = G(x) + C \ \forall x \in (a, b).}$
- But F and G are both continuous on $[a, b]$ so, taking limits as $x \to a^+$ and $x \to b^-$, then:

$$F(a) = \lim_{x \to a^+} F(x) = \lim_{x \to a^+} \Big(G(x) + C \Big) = G(a) + C$$

$$\boxed{\Rightarrow F(a) = G(a) + C \text{ and } F(b) = G(b) + C.}$$

Hence, $\boxed{F(x) = G(x) + C \ \forall x \in [a, b] \Big(* \Big).}$

- $G(a) = \int_a^a f(t)\, dt = 0$
  - Use $(*)$ then

$$\boxed{\Rightarrow F(b) - F(a) = \Big[G(b) + C\Big] - \Big[G(a) + C\Big]}$$

$$= G(b) - \underbrace{G(a)}_{=0}$$

$$= G(b) \quad \boxed{= \int_a^b f(t)\, dt \checkmark}$$

Example 4. Find $F'(x)$ for $\boxed{F(x) = \int_1^{x^4} \frac{1}{\cos(t)}\, dt.}$

Solution. Let $\boxed{G(u) = \int_1^u \frac{1}{\cos(t)}\, dt}$ and $\boxed{u = x^4 \Rightarrow \frac{du}{dx} = 4x^3}$, so use chain rule:

$$\boxed{F'(x) = \frac{d}{dx}\left[\int_1^{x^4} \frac{1}{\cos(t)}\, dt\right] = \frac{d}{dx} G(u(x)) = G'(u(x))u'(x)}$$

$$= \frac{d}{du}\left[\int_1^{u} \frac{1}{\cos(t)}\, dt\right]\frac{du}{dx} = \frac{1}{\cos(u)} 4x^3 \boxed{= \frac{1}{\cos(x^4)} \cdot 4x^3 \ \checkmark}$$

The Fundamental Theorem of Calculus for a composition

Let f be continuous and g differentiable, then

$$\frac{d}{dx}\left[\int_a^{g(x)} f(t)\,dt\right] = f(g(x)) \cdot g'(x)$$

Proof. Let $\boxed{u = g(x) \Rightarrow \frac{du}{dx} = g'(x)}$, use chain rule:

$$\boxed{F'(x) = \frac{d}{dx}\left[\int_a^{g(x)} f(t)\,dt\right]} = \frac{d}{du}\left[\int_a^{u} f(t)\,dt\right]\frac{du}{dx}$$

$$= f(u) \cdot g'(x)$$

$$\boxed{= f(g(x)) \cdot g'(x) \checkmark}$$

The Fundamental Theorem of Calculus for a composition

$$\frac{d}{dx}\left[\int_a^{g(x)} f(t)\,dt\right] = f(g(x)) \cdot g'(x)$$

Example 5. compute F' if $F(x) = \int_{\sqrt{x}}^{2(x+1)^3} \sin(t)\,dt$.

Solution. $F'(x) = \frac{d}{dx}\left[\int_{\sqrt{x}}^{2(x+1)^3} \sin(t)\,dt\right]$

$$= \frac{d}{dx}\left[\int_{\sqrt{x}}^{1} \sin(t)\,dt + \int_{1}^{2(x+1)^3} \sin(t)\,dt\right]$$

$$= \frac{d}{dx}\left[-\int_{1}^{\sqrt{x}} \sin(t)\,dt\right] + \frac{d}{dx}\left[\int_{1}^{2(x+1)^3} \sin(t)\,dt\right]$$

$$= -\sin(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}} + \sin(2(x+1)^3) \cdot 6(x+1)^2 \checkmark$$

The Fundamental Theorem of Calculus for a composition

Corollary:

$$\frac{d}{dx}\left[\int_{g(x)}^{h(x)} f(t)\, dt\right] = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x)$$

Proof. Splitting by any value $x = c$ on the domain of f, the given integral may be rewritten as a sum of two terms, for which we can apply the theorem:

$$\boxed{\int_{g(x)}^{h(x)} f(t)\, dt} = \int_{g(x)}^{c} f(t)\, dt + \int_{c}^{h(x)} f(t)\, dt = \boxed{-\int_{c}^{g(x)} f(t)\, dt + \int_{c}^{h(x)} f(t)\, dt}$$

Then, the derivative takes the following form

$$\boxed{\frac{d}{dx}\left[\int_{g(x)}^{h(x)} f(t)\, dt\right] = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x) \checkmark}$$

The Fundamental Theorem of Calculus

Example 6. Find the intervals of increase and decrease and the local minimum and maximum values of $\boxed{F(x) = \int_{-2}^{x} t\, e^{t^2+3}\, dt}$ on $(-2, \infty)$.

Solution. • Since $f(t) = t\, e^{t^2+3}$ is continuous on $[-2, \infty)$, we may apply the theorem.

• $F'(x) = \frac{d}{dx}\left[\int_{-2}^{x} t\, e^{t^2+3}\, dt\right] = x\, e^{x^2+3}$ on $(-2, \infty)$

	$(-2, 0)$	0	$(0, \infty)$
$F'(x) = xe^{x^2+3}$	$-$	0	$+$
F	*decreasing*	local minimum	*increasing*

Local minimum, no maximum

- F has no local maximum on $(-2, \infty)$
- F attains a local minimum at $x = 0$ with value

$$\boxed{F(0) = \int_{-2}^{0} t\, e^{t^2+3}\, dt = \frac{1}{2} e^{t^2+3}\Big|_{t=-2}^{t=0} = \frac{1}{2}(e^3 - e^7) \checkmark}$$

since

$$\frac{d}{dt}\left[\frac{1}{2} \cdot e^{t^2+3}\right] = t \cdot e^{t^2+3} \Rightarrow F(t) = \frac{1}{2} e^{t^2+3} = \int t \cdot e^{t^2+3} dt$$

The Substitution Rule for Definite Integrals

Let f be continuous on $[a,b]$ and g differentiable on (a,b), then $\int_a^b f(g(x)) \cdot g'(x) = \int_{g(a)}^{g(b)} f(u)\, du$

Proof. Let F be any antiderivative of f and $\boxed{u = g(x) \Rightarrow \frac{du}{dx} = g'(x)}$, then by the Fundamental Theorem, we have:

- $\boxed{\int_a^b f(g(x)) \cdot g'(x)\, dx = F(g(x))\Big|_a^b = F(g(b)) - F(g(a))}$

- Also, $\int_{g(a)}^{g(b)} f(u)\, du = F(u)\Big|_{(g(a)}^{g(b)} = F(g(b)) - F(g(a))$ ✓

Example 7. Compute

$$\int_0^4 \sqrt{2x+1}\,dx$$

Solution 1) Find the antiderivative and then evaluate the definite integral: we start with the substitution $u = 2x+1$, then

$$\int \sqrt{2x+1}\,dx = \int \sqrt{u}\,\frac{du}{2} = \frac{1}{2}\frac{2}{3}u^{3/2} = \frac{1}{3}u^{3/2}$$

$$= \frac{1}{3}(2x+1)^{3/2}$$

Then evalute the definite integral using the previous antiderivative:

$$\int_0^4 \sqrt{2x+1}\,dx = = \frac{1}{3}(2x+1)^{3/2}\Big|_{x=0}^{x=4} = \frac{1}{3}\left[9^{3/2} - 1^{3/2}\right] = \frac{26}{3}$$

Solution 2) Use the Substituion Rule for the definite integral: Substitute

- $g(x) = u = 2x + 1$, then $g(0) = 1$ and $g(4) = 9$,
- $g'(x)dx = 2dx = du$ and $dx = \frac{du}{2}$, then

$$\int_0^4 \sqrt{2x+1}\, dx$$

$$= \int_a^b f(g(x)) \cdot g'(x)dx = \int_{g(a)}^{g(b)} f(u)\, \frac{du}{2}$$

$$= \int_1^9 \frac{1}{2}\sqrt{u}\, du = \frac{1}{3}u^{3/2}\Big|_{u=1}^{u=9}$$

$$= \frac{1}{3}\left[9^{3/2} - 1^{3/2}\right] = \frac{26}{3} \checkmark$$

The Substitution Rule for Definite Integrals

Example 8. Compute

$$\int_1^e \left[\frac{\ln^2(x)}{x} + \frac{1}{(3-5x)^2}\right]\, dx$$

Solution. Write the given integral as a sum of two more simple ones:

$$\int_1^e \left[\frac{\ln^2(x)}{x} + \frac{1}{(3-5x)^2}\right]\, dx$$

$$= \int_1^e \frac{\ln^2(x)}{x}\, dx + \int_1^e \frac{1}{(3-5x)^2}\, dx = A + B$$

Compute A= $\int_1^e \frac{\ln^2(x)}{x}\,dx$ using substitution $u = \ln(x)$ so $du = \frac{1}{x}dx$ and $\ln(1) = 0,\ \ln(e) = 1$:

$$\boxed{\int_1^e \frac{\ln^2(x)}{x}\,dx = \int_0^1 u^2}$$

$$= \frac{1}{3}u^3\Big|_{u=0}^{u=1} = \frac{1}{3} \checkmark$$

The Substitution Rule for Definite Integrals

Compute B= $\displaystyle\int_1^e \frac{1}{(3-5x)^2}\,dx$ using a different susbtitution:

$g(x) = u = (3-5x) \Rightarrow du = -5\,dx \Rightarrow dx = -\frac{1}{5}\,du$, then

$$\boxed{g(1) = -2,\ g(e) = 3-5e}$$

Then

$$\boxed{\int_1^e \frac{1}{(3-5x)^2}\,dx = -\frac{1}{5}\int_{-2}^{3-5e} u^{-2}\,du}$$

$$= +\frac{1}{5}u^{-1}\Big|_{u=-2}^{3-5e} = \frac{1}{5}\Big[-\left(\frac{1}{3-5e}\right)+\frac{1}{2}\Big] \checkmark$$

Hence, the given integral is the sum of the two, A+B:

$$\int_1^e \left[\frac{\ln^2(x)}{x} + \frac{1}{(3-5x)^2}\right] dx = A + B$$

$$= \frac{1}{3} + \frac{1}{5}\left[-\left(\frac{1}{3-5e}\right) + \frac{1}{2}\right]$$

$$\boxed{= \frac{13}{30} + \frac{1}{5}\left(\frac{1}{5e-3}\right) \checkmark}$$